# Achievers — Level 2 — Answer Key

1. **C) 48** — Working backwards: (27 − 15) × 4 = 12 × 4 = 48.
2. **C) 70** — Let the first number be $x$. Then second = $3x$, third = $3x-5$. Sum: $x + 3x + (3x-5) = 170 \rightarrow 7x = 175 \rightarrow x = 25$. Third = $3\times25-5 = 70$.
3. **C) ₹1,680** — After spending 3/7 on toys, 4/7 of the money remains. After spending 1/4 of that on books, 3/4 of 4/7 = 3/7 of the original remains. So 3/7 of the total = ₹720 → total = ₹1,680.
4. **C) 10** — Let the son's age be $x$; father's age = $4x$. After 5 years: $4x+5 = 3(x+5) \rightarrow 4x+5 = 3x+15 \rightarrow x = 10$.
5. **B) 4 hours** — Pipe A fills 1/6 of the tank per hour, Pipe B fills 1/12 per hour. Together: 1/6 + 1/12 = 3/12 = 1/4 of the tank per hour, so the tank fills in 4 hours.
6. **C) 32** — Let the numbers be $x$, $x+2$, $x+4$. Sum: $3x+6 = 90 \rightarrow x = 28$. Largest = 28+4 = 32.
7. **C) 50** — Larger number = (sum + difference) ÷ 2 = (86+14) ÷ 2 = 50.
8. **C) 11** — Let $a$ = number of ₹5 coins, $b$ = number of ₹2 coins. $a+b = 18$ and $5a+2b = 69$. Substituting $b = 18-a$: $5a + 2(18-a) = 69 \rightarrow 3a + 36 = 69 \rightarrow a = 11$.
9. **B) 7 hours** — Speed = 180 ÷ 3 = 60 km/h. Time for 420 km = 420 ÷ 60 = 7 hours.
10. **C) 36** — Let the tens digit be $t$; the ones digit is $t+3$. The number = $11t+3$, and the digit sum = $2t+3$. Setting $11t+3 = 4(2t+3)$ gives $11t+3 = 8t+12 \rightarrow t = 3$. Number = 11×3+3 = 36. Check: digit sum = 3+6 = 9, and 4×9 = 36. ✓
11. **C) 176 sq cm** — Perimeter = $2(l+b) = 54 \rightarrow l+b = 27$. Since $l = b+5$: $2b+5 = 27 \rightarrow b = 11$, $l = 16$. Area = 11×16 = 176 sq cm.
12. **C) 14** — Each term = (sum of previous two) − 1: after 2, 3, 4, 6, 9, the next term is 6+9−1 = 14.
13. **C) 120** — Let the original number of pens be $x$. After Monday, 2/3 of $x$ remain. After Tuesday: $2/3x - 24 = 56 \rightarrow 2/3x = 80 \rightarrow x = 120$.
14. **C) 10:00 a.m.** — The bells next ring together after the LCM of 12, 15, and 20 minutes, which is 60 minutes. 9:00 a.m. + 60 min = 10:00 a.m.
15. **C) 83** — The number must leave remainder 3 when divided by 8 and remainder 2 when divided by 9. Checking numbers between 50 and 100 that fit both conditions gives 83 (83 ÷ 8 = 10 remainder 3; 83 ÷ 9 = 9 remainder 2).

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**How to use this:** 10+/15 is strong for a Level-2-focused set. For anything missed, redo it without looking at the answer key first — these question types (working backwards, combining fractions-of-remainder, rate/work problems) repeat constantly in the Achievers section of real papers.